

Matteo Palmonari

matteo.palmonari@disco.unimib.it

5.4

Information Extraction: Interpretation and Annotation of Tabular Data



dat·AI

data * AI
AI * data

Tabular Data Interpretation

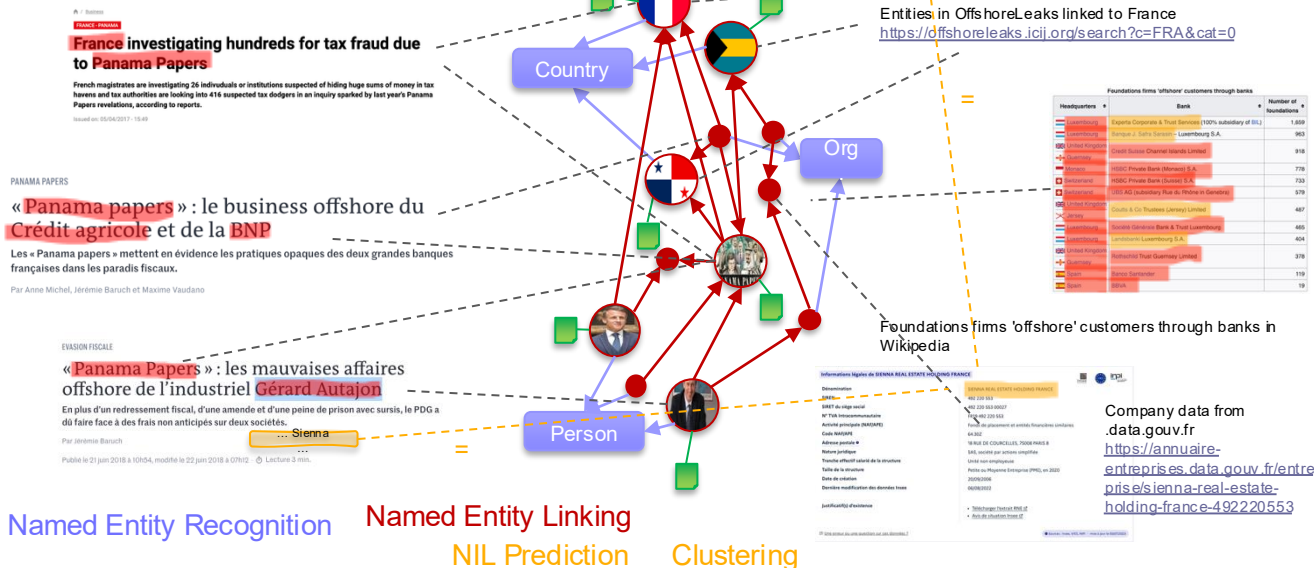
Tables contain a vast amount of data

- the Common Crawl* repository contains **230 million web tables** [Lehmberg et al. 2016]
- the current snapshot of **Wikipedia** contains more than **3.23 million tables** [Fetahu et al. 2019]
- the number of worldwide **spreadsheet** users is estimated to exceed 400 million, and about 50 to 80% of businesses use spreadsheets
- a very large amount of data manipulated by companies is in tabular format...

* <http://commoncrawl.org/>

Reminder: Semantic Data Integration

Annotations: data linking



Reminder: Tabular Data Enrichment

Classify the German states in the State column as belonging to "east" or "west" Germany. Output only "east" or "west"

Linking to KGs or other data sources using reconciliation algorithms

Extension by fetching data from KGs

Extension by inference using a model



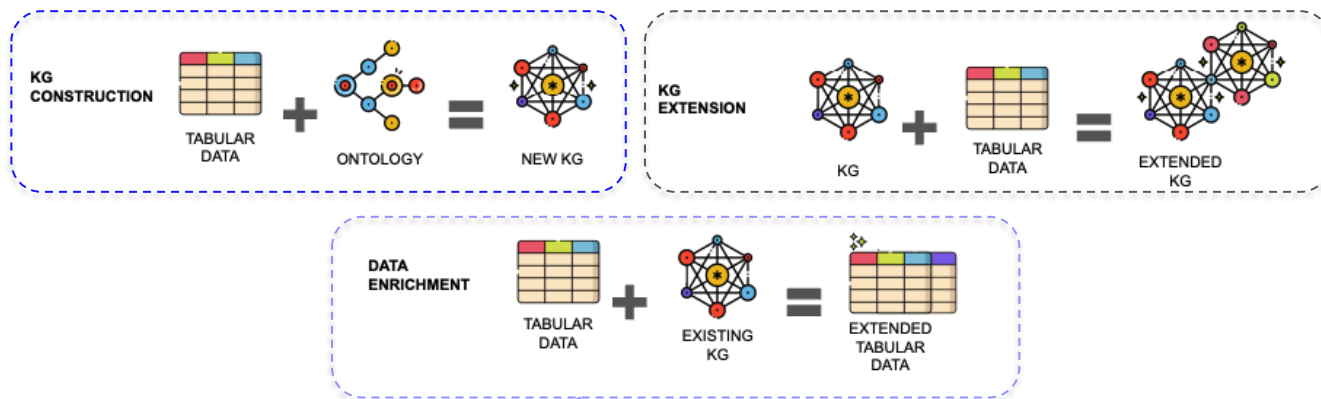
Keyword	#im	City	State	ID (Geonames)	Coordinates (Geonames)	ID (Wikidata)	Population (Wikidata)	Temp (OpenW)	Date	Area
194906	64	Altenburg	Thuringia	2822542	[50.98763, 12.43684]	Q1205	2,122,335	18°	2017-03-11	East
517827	50	Inglostadt	Bavaria	2951839	[48.76508, 11.42372]	Q980	13,435,062	17°	2017-03-12	West
459143	42	Berlin	Berlin	2950157	[52.52437, 13.41053]	Q648102	3,782,202	17°	2017-03-12	East
891139	36	Munich	Bavaria	2951839	[48.13743, 11.57549]	Q980	13,435,062	19°	2017-03-11	West
459143	30	Nuremberg	Bavaria	2951839	[49.45421, 11.07752]	Q980	13,435,062	12°	2017-03-12	East

here



Extension by fetching data from other data sources via APIs

Downstream Applications: Summary



Keyword	#im	City	Region	ID (Geonames)	Latitude (Geonames)	Longitude (Geonames)	ID (Wikidata)	Area (Wikidata)	Temp (ECMWF)	Date
194906	64	Altenburg	Thuringia	2822542	50.98763	12.43684	Q1205	45.6 km ²	18°	11/03/2017
517827	50	Inglostadt	Bavaria	2951839	48.76508	11.42372	Q980	133.35 km ²	17°	12/03/2017
459143	42	Berlin	Berlin	2950157	52.52437	13.41053	Q648102	891.68 km ²	17°	12/03/2017
891139	36	Munich	Bavaria	2951839	48.13743	11.57549	Q980	310.71 km ²	19°	11/03/2017
459143										0/03/2017

Example of data enrichment by composing different individual linking and extension services

Data enrichment by the link & extend paradigm

What is This Table Describing?

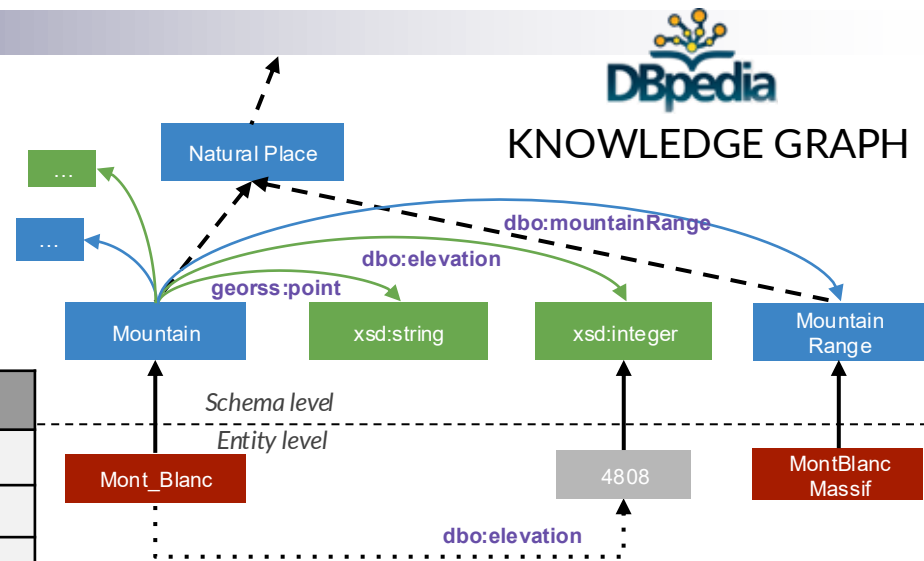
Mount Everest	8,848	Himalayas	May 29, 1953
K-2 (Godwin Austin)	8,611	Karakoram	July 31, 1954
Kanchenjunga	8,597	Himalayas	May 25, 1955
Lhotse	8,511	Himalayas	May 18, 1956
Makalu I	8,481	Himalayas	May 15, 1955
Dhaulagiri I	8,167	Himalayas	May 13, 1960
Manaslu	8,156	Himalayas	May 9, 1956
Cho Uyo	8,153	Himalayas	Oct 19, 1954
Nanga Parbat	8,124	Himalayas	July 3, 1953
Annapurna I	8,078	Himalayas	June 3, 1950
Gasherbrum I	8,068	Karakoram	July 5, 1958
Broad Peak I	8,047	Karakoram	June 9, 1957
Gasherbrum II	8,034	Karakoram	July 7, 1956
Shisha Pangma (Gasainthan)	8,013	Himalayas	May 2, 1964
Gasherbrum III	7,952	Karakoram	Aug 11, 1975
Annapurna II	7,937	Himalayas	May 17, 1960
Gasherbrum IV	7,923	Karakoram	Aug 6, 1958

What is This Table Describing?

MOUNTAIN	HEIGHT IN METERS	RANGE	CONQUERED ON
Mount Everest	8,848	Himalayas	May 29, 1953
K-2 (Godwin Austin)	8,611	Karakoram	July 31, 1954
Kanchenjunga	8,597	Himalayas	May 25, 1955
Lhotse	8,511	Himalayas	May 18, 1956
Makalu I	8,481	Himalayas	May 15, 1955
Dhaulagiri I	8,167	Himalayas	May 13, 1960
Manaslu	8,156	Himalayas	May 9, 1956
Cho Uyo	8,153	Himalayas	Oct 19, 1954
Nanga Parbat	8,124	Himalayas	July 3, 1953
Annapurna I	8,078	Himalayas	June 3, 1950
Gasherbrum I	8,068	Karakoram	July 5, 1958
Broad Peak I	8,047	Karakoram	June 9, 1957
Gasherbrum II	8,034	Karakoram	July 7, 1956
Shisha Pangma (Gasainthan)	8,013	Himalayas	May 2, 1964
Gasherbrum III	7,952	Karakoram	Aug 11, 1975
Annapurna II	7,937	Himalayas	May 17, 1960
Gasherbrum IV	7,923	Karakoram	Aug 6, 1958

Tabular Data Annotation

Name	Coordinates	Height	Range
Le Mont Blanc	45°49'57"N 06°51'52"E	4808	M. Blanc massif
Hohtälli	45°98'96"N 07°80'25"E	3275	Pennine Alps
Monte Cervino	45°58'35"N 07°39'31"E	4478	Pennine Alps



Given

- a relational table T
- a Knowledge Graph (entities + statements) and an ontology (types + predicates)

T is annotated when:

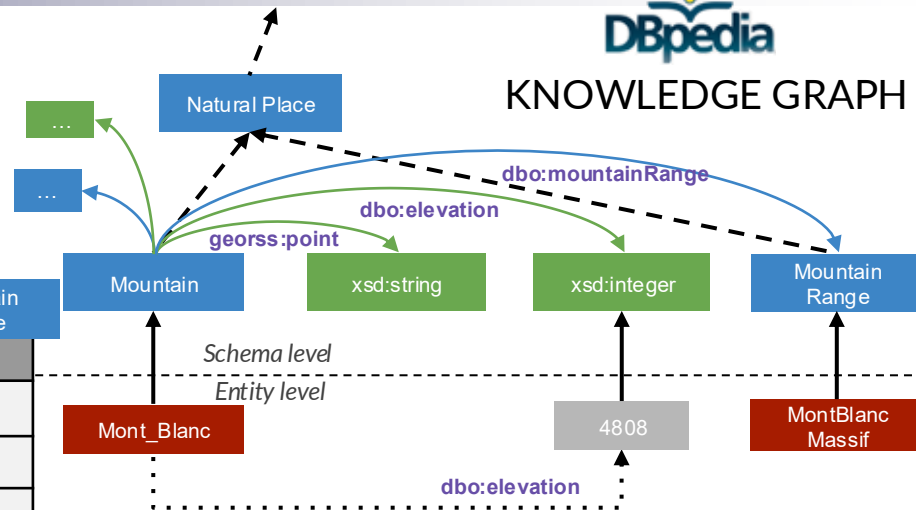
-
-
-

Tabular Data Annotation



KNOWLEDGE GRAPH

Mountain	xsd:string	xsd:integer	Mountain Range
Name	Coordinates	Height	Range
Le Mont Blanc	45°49'57"N 06°51'52"E	4808	M. Blanc massif
Hohtälli	45°98'96"N 07°80'25"E	3275	Pennine Alps
Monte Cervino	45°58'35"N 07°39'31"E	4478	Pennine Alps



Given

- a relational table T
- a **Knowledge Graph (entities + statements)** and an ontology (**types + predicates**)

T is annotated when:

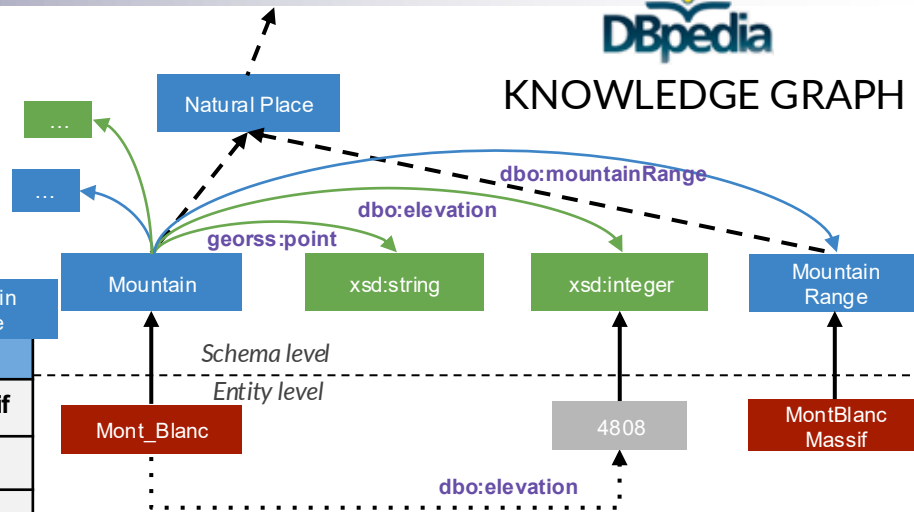
- each column is associated with one or more **KG-types** (CTA)
-
-

Tabular Data Annotation



KNOWLEDGE GRAPH

Mountain	xsd:string	xsd:integer	Mountain Range
Name	Coordinates	Height	Range
Mont_Blanc	45°49'57"N 06°51'52"E	4808	MontBlanc Massif
Hohtälli	45°98'96"N 07°80'25"E	3275	Pennine Alps
Monte Cervino	45°58'35"N 07°39'31"E	4478	Pennine Alps



Given

- a relational table T
- a **Knowledge Graph** (entities + statements) and an ontology (types + predicates)

T is annotated when:

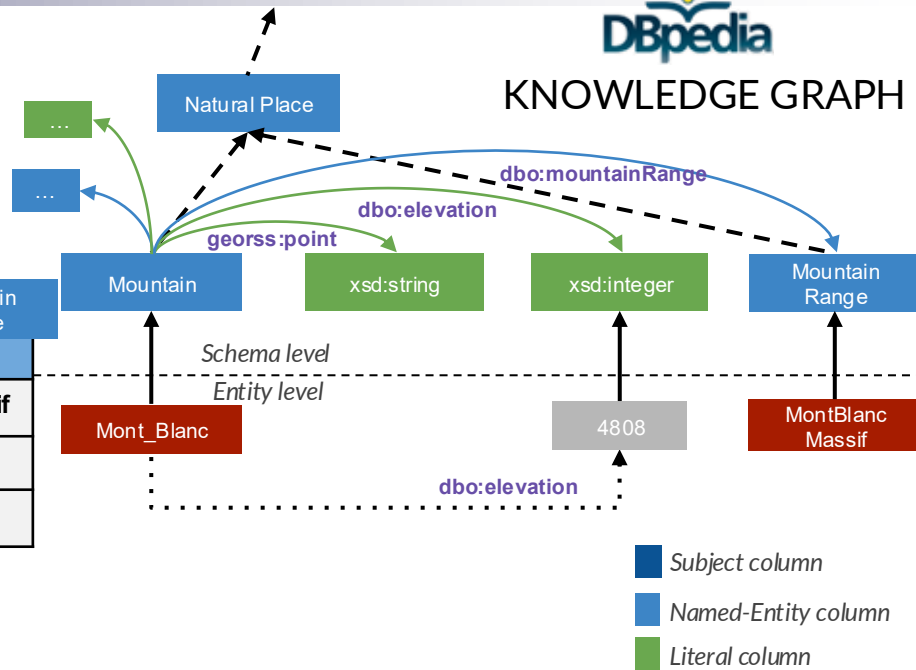
- each column is associated with one or more **KG-types** (CTA)
- each cell in “entity columns” is annotated with a **KG-entity** (CEA)
-

Tabular Data Annotation



KNOWLEDGE GRAPH

Mountain	xsd:string	xsd:integer	Mountain Range
Name	Coordinates	Height	Range
Mont_Blanc	45°49'57"N 06°51'52"E	4808	MontBlanc Massif
Hohtälli	45°98'96"N 07°80'25"E	3275	Pennine Alps
Monte Cervino	45°58'35"N 07°39'31"E	4478	Pennine Alps



Given

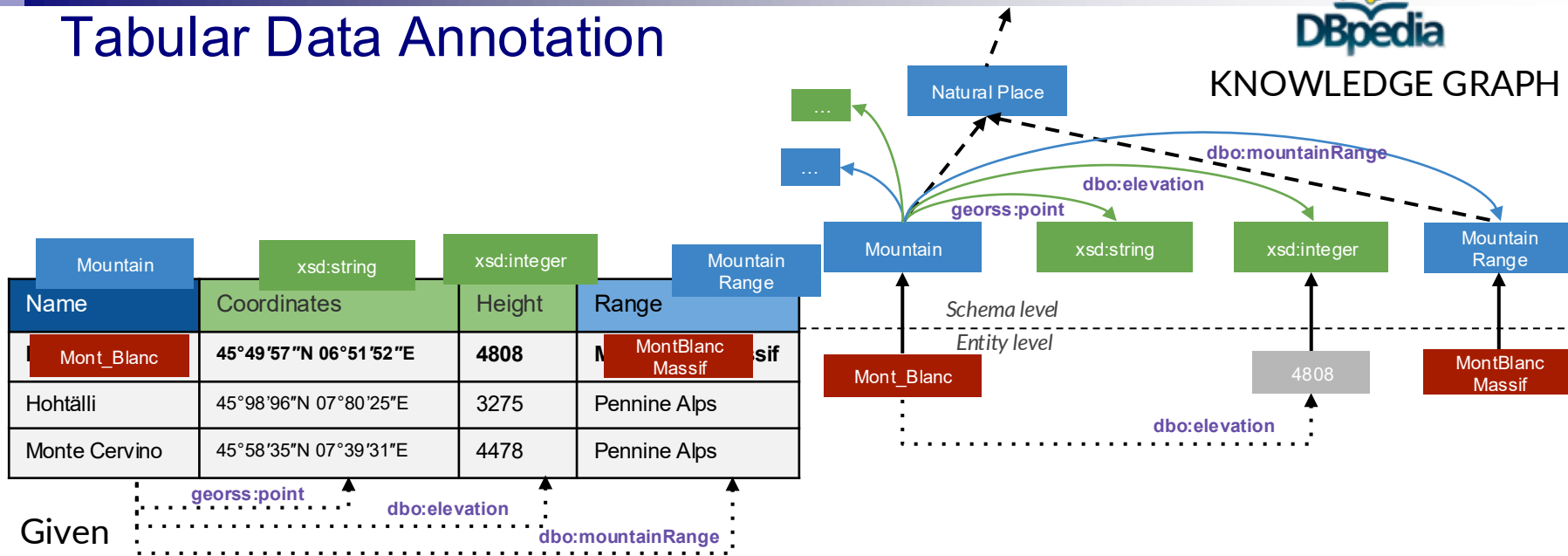
- a relational table T
- a Knowledge Graph (entities + statements) and an ontology (types + predicates)

T is annotated when:

- each column is associated with one or more **KG-types** (CTA)
- each cell in "entity columns" is annotated with a **KG-entity** (CEA)

Also referred to as "entity linking" (for tables)
... or RECONCILIATION

Tabular Data Annotation



- a relational table T
- a Knowledge Graph (entities + statements) and an ontology (types + predicates)

T is annotated when:

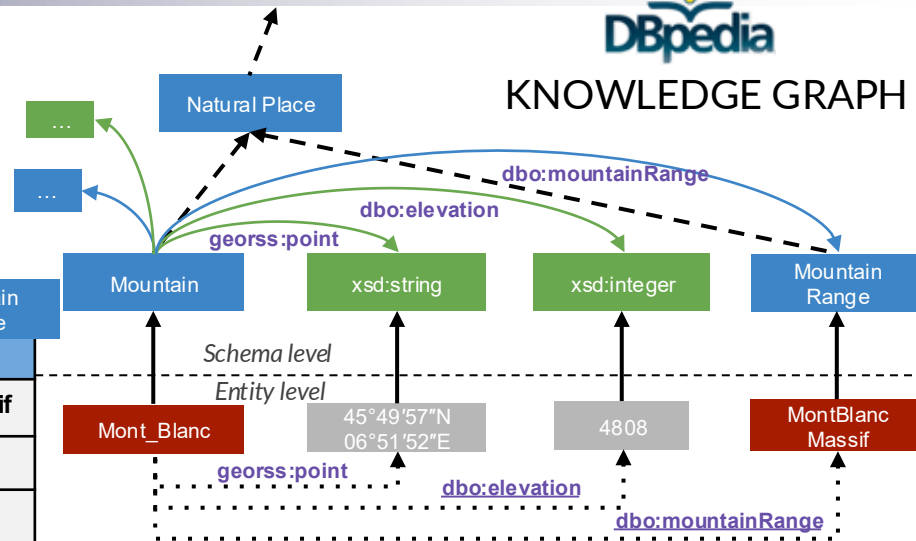
- each column is associated with one or more **KG-types** (CTA)
- each cell in "entity columns" is annotated with a **KG-entity** (CEA)
- some pair of columns is annotated with a binary **KG-predicate** (CPA)

Tabular Data Annotation ... for KG completion



KNOWLEDGE GRAPH

Mountain	xsd:string	xsd:integer	Mountain Range
Name	Coordinates	Height	Range
Mont_Blanc	45°49'57"N 06°51'52"E	4808	MontBlanc Massif
Hohtälli	45°98'96"N 07°80'25"E	3275	Pennine Alps
Monte Cervino	45°58'35"N 07°39'31"E	4478	Pennine Alps



Given



- a relational table T
- a Knowledge Graph (entities + statements) and an ontology (types + predicates)

T is annotated when:

- each column is associated with one or more **KG-types** (CTA)
- each cell in "entity columns" is annotated with a **KG-entity** (CEA)
- some pair of columns is annotated with a binary **KG-predicate** (CPA)

Tabular Data Annotation ... with novel entities

Mountain	xsd:string	xsd:integer	Mountain Range
Name	Coordinates	Height	Range
Mont_Blanc	45°49'57"N 06°51'52"E	4808	MontBlanc Massif
[NIL: Hohtälli]	45°98'96"N 07°80'25"E	3275	Pennine Alps
Monte Cervino	45°58'35"N 07°39'31"E	4478	Pennine Alps

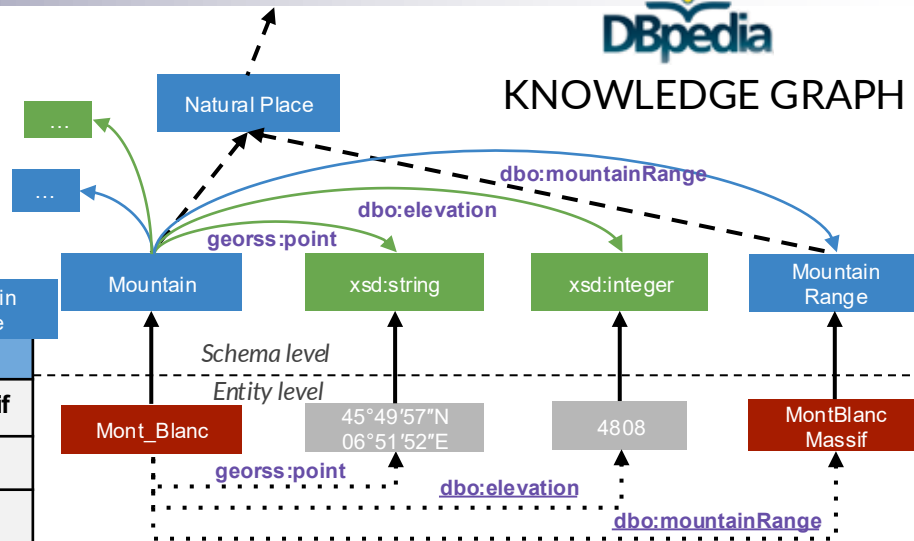
Given



- a relational table T
- a Knowledge Graph (entities + statements) and an ontology (types + predicates)

T is annotated when:

- each column is associated with one or more **KG-types** (CTA)
- each cell in “entity columns” is annotated with a **KG-entity** or with **NIL** (if not in the KG)
- some pair of columns is annotated with a binary **KG-predicate** (CPA)





TUTSTI

Marco Cremaschi*,
Fabio D'Adda
Ernesto Jimenez-Ruiz,
Matteo Palmonari

[*marco.cremaschi@unimib.it](mailto:marco.cremaschi@unimib.it)

November 12th 2024

Semantic Table Interpretation from Heuristic to LLM-based approaches



Resources & Acknowledgements

Key resources:

- **Website:** <https://unimib-datai.github.io/sti-website/tutorial/>
- **Hands-on session material:**
https://drive.google.com/drive/folders/1irbTMKvP4INm_8SdJne_jNSIHgegghBL
- **New survey:** <https://arxiv.org/abs/2411.11891>

Disclaimers:

- We keep citations light on these slides; please check the website and the survey for complete references

Partially
supported by



enRichMyData



For more updates check:

<https://enrichmydata.eu/>

<https://www.discountquality.polimi.it/>

This work presented in this presentation has received funding from the European Union's Horizon 2020 research and innovation program under grant agreements No 101070284 - enRichMyData and the Italian PRIN project Discount Quality for Responsible Data Science: Human-in- the-Loop for Quality Data (202248FWFS) funded by the European Community - Next Generation EU.



Example of heuristics approach

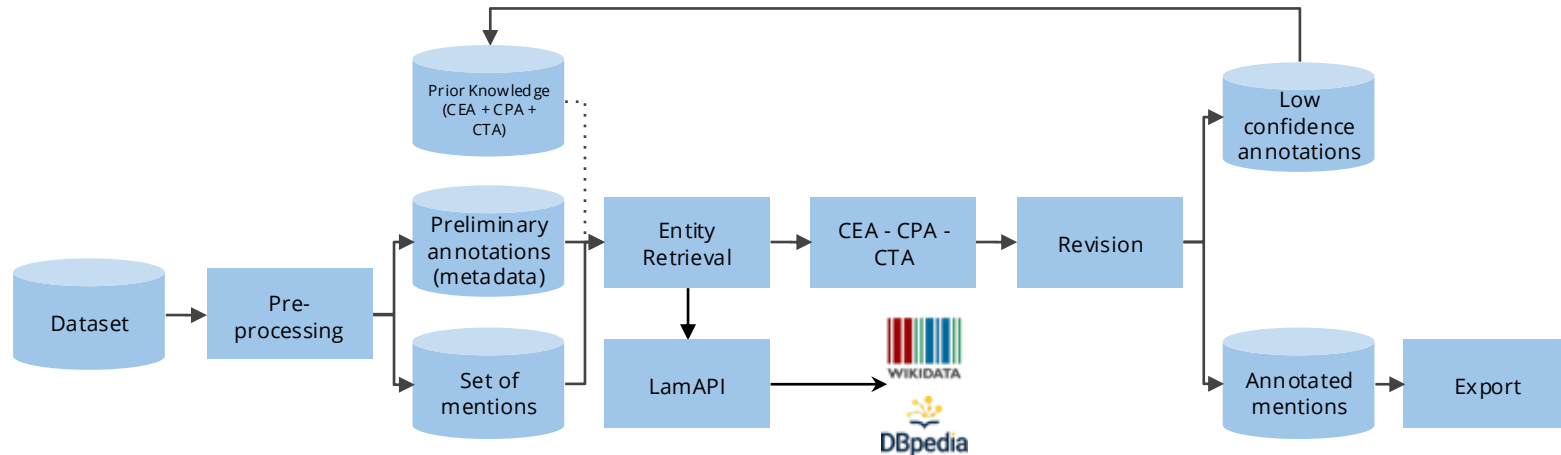
END-TO-END STI

Example of Heuristic STI Pipeline: s-elBat



SKIP

1. Preprocessing and Data preparation
2. Entity Retrieval
3. Cell Entity Annotation (CEA)
4. Cell Property Annotation (CPA)
5. Cell Type Annotation (CTA)
6. Revision
7. Export



From **Heuristics** to **Feature-based ML** Approaches

CEA / RECONCILIATION

W3C Reconciliation Service API

- A protocol designed for matching and linking named entities (like people, locations, or organizations) across different datasets on the web.
- Service-Oriented Approach:
 - Exposes endpoints that accept structured queries.
 - Returns potential matches with scores indicating the confidence of each match.
 - Supports standard HTTP GET/POST requests.
- Often used in applications like OpenRefine, it enables users to clean and align data by reconciling terms with canonical databases or knowledge graphs.

W3C Reconciliation Service API

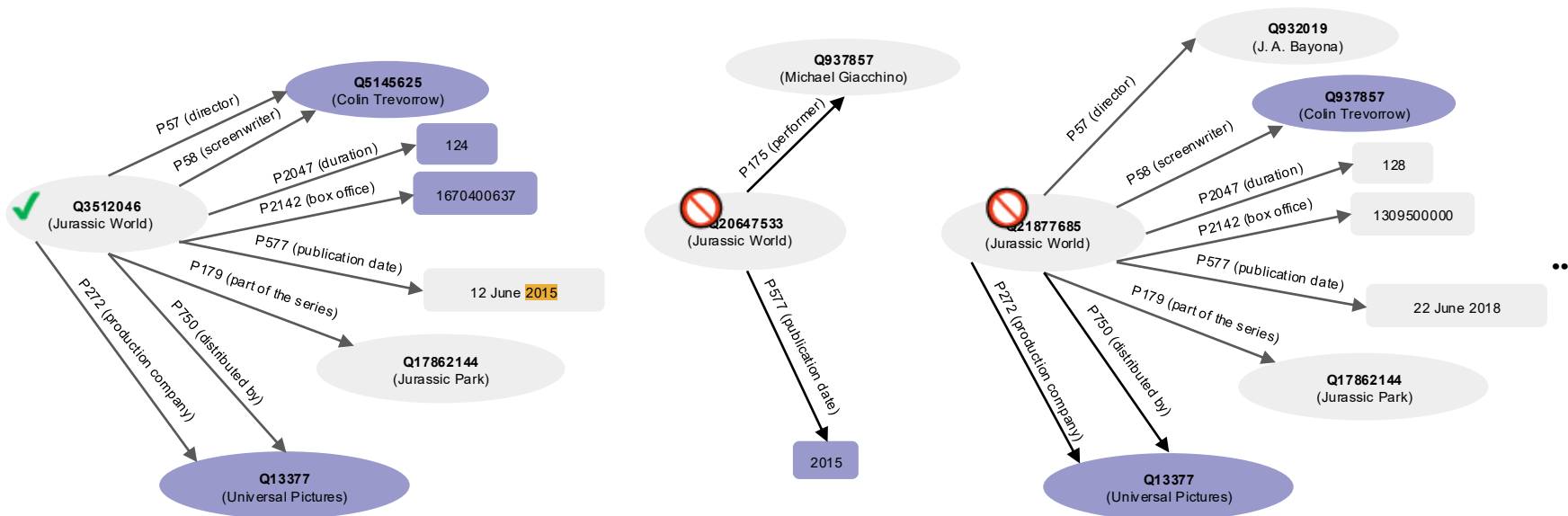
Query Structure

- Single query: query={"query":"entity name", "type":"/domain/type"}
- Multi-query: Encapsulates multiple entity queries in a single API call
- Fields:
 - query:** The search term.
 - type:** Entity type (e.g., /people/person).
 - properties:** Array of key-value pairs for additional filters.

Response Structure

- Returns a JSON object with:
 - id:** Identifier of the matched entity.
 - name:** Name of the entity.
 - type:** Array of types the entity belongs to.
 - score:** Confidence score (e.g., 0.0–1.0).
 - match:** Boolean indicating if it's a confident match.

Challenges: Entity Disambiguation and Ranking in Tables



title	director	release year	domestic distributor	length in min	worldwide gross
jurassic world	colin trevorrow	2015	universal pictures	124	1670400637

Generalistic approaches to **tabular data understanding**

CEA / RECONCILIATION

+ OTHER STI + OTHER TASKS

(L)LM-based: Generalistic

- **TURL**: fine-tuned TinyBERT for tabular data understanding
 - Generic model + models for specific tasks (task-specific fine-tuning)
 - Parameters: 4M; context: 512 tokens
- **TableLlama**: fine-tuned Llama for tabular data understanding and manipulation
 - Generic model with prompting
 - Parameters: Llama fine-tuned with LongLoRA 7B; context: 8k
- Also relevant:
 - **UNICORN** (multi-task cross-encoder, incl. CTA, CEA)
 - **TableGPT**, which performs CTA

(L)LM-based Approaches: Tasks Summary

	Tasks					
	Annotation/Matching	Augmentation	QA	Fact Verification	Dialogue Generation	Data-to-Text
Unicorn	CEA CTA Entity Matching Entity Alignment Ontology Matching Schema Matching String Matching	N.A.	N.A.	N.A.	N.A.	N.A.
TURL	CEA CTA CPA	Row Population Cell Filling Schema Aug.	N.A.	N.A.	N.A.	N.A.
TableLlama	CEA CTA CPA	Row Population Schema Aug.	Hierarchical Table QA Highlighted Cells QA Hybrid Table QA Table QA	Fact Verification	Table Grounded Dialogue Generation	Highlighted Cells Description

Interesting for data enrichment

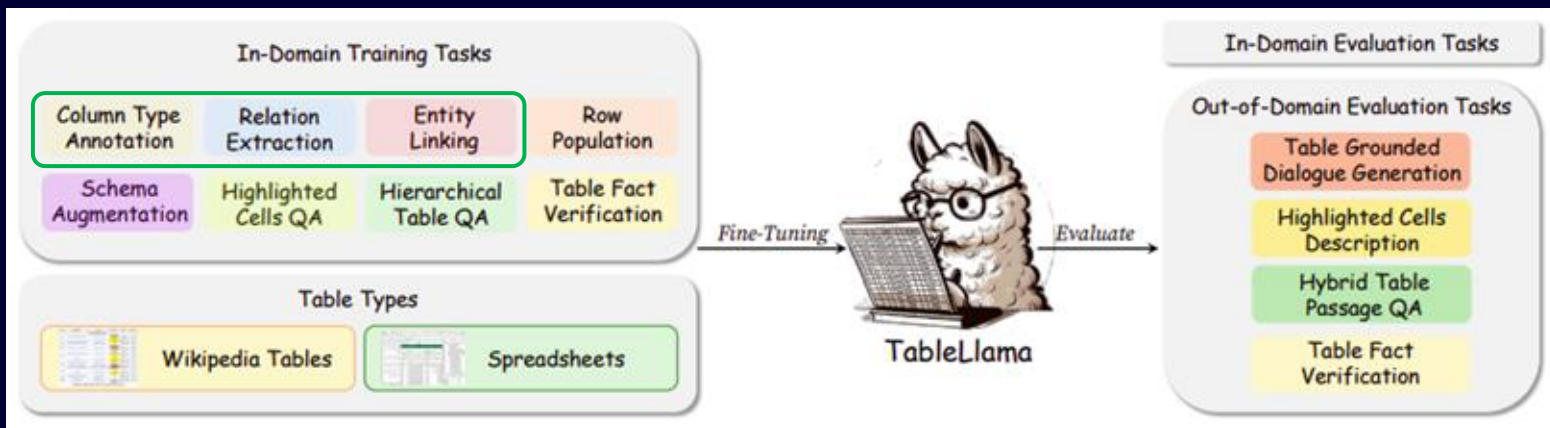
Interesting for interactive exploration

(L)LM-based Approaches: Inference Summary

	Input	Output	Transformer	Params
Unicorn	Encode pairs as: [CLS] S(a) [SEP] S(b) [SEP] where S(*) is a generic function for serializing any pair (a, b) from the matching tasks into a text sequence	A score in [0,1] for every pair (mention, i^{th} -candidate)	DeBERTa (Encoder-only)	147M
TURL	Flatten input table as: [Table caption, Table Header-1, ..., Table Header-M, Row-1, ..., Row-N]	A probability distribution over the N candidates for a mention	Tiny-BERT (Encoder-only)	14.5M
TableLlama	Prompt based: <instruction, table input, question> • <u>Instruction</u> is a detailed task description • <u>Table input</u> is the concatenation of table metadata (Wikipedia page title, section title and table caption) with the serialized table • <u>Question</u> contains all the information the model needs to complete the task and prompt it to generate an answer.	Autoregressively generated answer given the prompted question	Llama 2 (Decoder-only)	7B

TableLlama: Overview

STI tasks



Training with TableInstruct

- Dataset with 14 datasets for 11 tasks
- 1.24M tables

TableLlama: CTA example

Candidates are provided

Similar as in RAG, but
here we need to retrieve
candidate types

Column Type Annotation

Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately completes the request.

Instruction:

This is a column type annotation task. The goal for this task is to choose the correct types for one selected column of the table from the given candidates. The Wikipedia page, section and table caption (if any) provide important information for choosing the correct column types.

Input:

[TLE] The Wikipedia page is about 1958 Nippon Professional Baseball season. The Wikipedia section is about Central League. The table caption is Pitching leaders. [TAB] col: | stat | player | team | total | [SEP] row 1: | Wins | Masaichi Kaneda | Kokutetsu Swallows | 31 | [SEP] row 2: | Losses | Noboru Akiyama | ...

Question:

The column 'player' contains the following entities: <Masaichi Kaneda>, <Noboru Akiyama>, etc. The column type candidates are: tv.tv_producer, astronomy.star_system_body, location.citytown, sports.pro_athlete, biology.organism, medicine.muscle, baseball.baseball_team, baseball.baseball_player, aviation.aircraft_owner, people.person, ... What are the correct column types for this column (column name: player; entities: <Masaichi Kaneda>, <Noboru Akiyama>, etc)?

Response:

sports.pro_athlete, baseball.baseball_player, people.person.

TableLlama: CEA example

Candidates are provided

Entity Linking

Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately completes the request.

Instruction: This is an entity linking task. The goal for this task is to link the selected entity mention in the table cells to the entity in the knowledge base. You will be given a list of referent entities, with each one composed of an entity name, its description and its type. Please choose the correct one from the referent entity candidates. Note that the Wikipedia page, Wikipedia section and table caption (if any) provide important information for choosing the correct referent entity.

Input: [TLE] The Wikipedia page is about A-League all-time records. The Wikipedia section is about Average season attendances. [TAB] col: | season | league average | total gate receipts | highest club | average | lowest club | average | row 1: | 2005-06 | 10,955 | 920,219 | Sydney FC | 16,669 | New Zealand Knights | 3,909 | [SEP] row 2: | 2006-07 | 12,927 | ...

Question: The selected entity mention in the table cell is: Melbourne Victory. The column name for 'Melbourne Victory' is highest club. The referent entity candidates are: <Melbourne Victory FC W-League [DESCRIPTION] None [TYPE] SoccerClub>, <2016\u20132017 Melbourne Victory FC season [DESCRIPTION] None [TYPE] SoccerClubSeason>, <2011\u20132012 Melbourne Victory season [DESCRIPTION] Association football club 2011/12 season for Melbourne Victory [TYPE] SoccerClubSeason>, ... What is the correct referent entity for the entity mention 'Melbourne Victory' ?

Response: <Melbourne Victory [DESCRIPTION] association football team from Australia [TYPE] SoccerClub>.

Similar as in RAG, but here we need to retrieve candidate entities (~20-50 from millions of entities).

Entity retrieval is a key step. We use LamAPI ([website](#)), a highly engineered retriever for Wikidata / DBpedia / Wikipedia

Where We're at Today? #1

- Recent paper evaluates CEA under different conditions on in-domain, moderately out-of-domain and out-of-domain data [\[Belotti & al. 2026\]](#)
 - ML-based approaches, e.g., Alligator [\[Avogadro & al. 2023\]](#) may still outperform LLM-based approaches in some circumstances, e.g., if properly fine-tuned on in-domain data
 - LLM-based approaches typically generalize better and seems to be the best solution today, especially for data from unknown distributions
 - Top-tier LLMs may be too slow and/or too expensive; but off-the-shelves models like GPT 4o-mini find a sweet spot in the performance/cost trade-off
 - Approaches fine-tuned on different tabular data tasks, e.g. TableLlama 7B, also provide good performance/cost trade-offs on tabular data tasks, including semantic annotation.
- Apparently, we need a ~128B 72B generalistic OS LLM to match TableLlama performance on CEA [\[Source: MammoTab Leaderboard\]](#)[\[Cremaschi et al. 2025\]](#) – it's Qwen2.5-72B-instruct, and was explicitly trained to better process tables and return JSON output

Where We're at Today? #2

- Some pragmatic aspects

- Schema-level annotations – CTA/CPA do not usually require intensive processing: top-tier LLMs are likely the best choice. They can be fully revised by a user if she wants a quality mapping
- Instance-level annotations – CEA require intensive processing: efficiency is an issue. Typically, they cannot be fully revised by a user, but in many cases they may want to revise most uncertain links

- Quantifying uncertainty: an intrinsic limitation of LLM-based approaches

- Uncertainty is key information to support HITL approaches with match revision – matching score is also part of W3C Reconciliation APIs
- LLM-based approaches do not natively estimate uncertainty
- We proposed an approach to estimate uncertainty for LLM-based data linking on tabular data [[Bono et al. 2025](#)] – uncertainty estimation for LLM is a cool and important topic

- In the enRichMyData project, we developed

- **W3C-compatible reconciliation services** wrapping ML-based approaches (e.g., Alligator) and LLM-based approaches
- Tools that use CEA – but also full STI (incl. CTA and CPA) in interactive applications supporting tabular data enrichment and KG construction (STI annotations → RDF files): especially **SemT-X** ([website](#))

SemT-X – Interactive Semantic Enrichment of Tabular Data

■ UI accessing external services

- STI (full)
- Reconciliation/linking services (W3C-compliant APIs)

- Geonames
- Wikidata
- Dbpedia
- ...

□ Extension services

- Wikidata (SPARQL)
- Weather extension
- HERE (georeferencing)
- Shortest-route
- ...

Tutorial video:

<https://www.youtube.com/watch?v=XMfRQueX48M>

Point of interest	Named Entity	Subject	type	location	inception	owned by	Place	Geonames ID	Adm1	Country	Foundation date
John F. Kennedy Presidential Library and Museum	Wikidata	Wikidata	type	location	inception	owned by	Place	Geonames ID	Adm1	Country	Foundation date
John F. Kennedy Presidential Library and Museum	Wikidata	Wikidata	type	location	inception	owned by	Place	Geonames ID	Adm1	Country	Foundation date
Petrie Museum of Egyptian Archaeology	Wikidata	Wikidata	type	location	inception	owned by	Place	Geonames ID	Adm1	Country	Foundation date
Helsinki City Museum	Wikidata	Wikidata	type	location	inception	owned by	Place	Geonames ID	Adm1	Country	Foundation date

Support to *Linking – Revision – Extension* of tabular data

■ Graphical view & revision of annotations

- Global and specific annotation rendering
- Single cell editing / annotation revision
- Column annotation revision

Upgrades

Two interfaces:

- UI
- Jupyter notebook (python)

All use SemT-X operations and interoperate on the same code. Once generated enrichment pipelines can be deployed to scale

Compliance-related information

Integration with LLMs

Next interface? ChatBot / LLM

More to do with LLM-integration

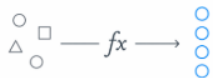
SemT-X

A framework for the semantic enrichment of tabular data by linking cells and columns to external knowledge sources, adding context and additional information.

Main Features

Modification

Prepare your data for enrichment by applying transformation functions.



Reconciliation

Link your table cells to external Knowledge Bases, such as Wikidata or GeoNames.



Matching Revision

Validate and correct candidate entities at cell level or through batch refinement and propagation across columns.



Extension

Enrich your table by fetching additional properties and metadata from reconciled entities.



Generative AI

Leverage Large Language Models to assist in AI-driven modification, reconciliation, and extension.



Automatic Annotation

Automatically annotate either the full table (Semantic Table Interpretation) or classify the column schema (NER).



Compliance Check

Evaluate the current state of your table's compliance against different privacy standards and legal regulations.



Visualization

Explore your data through interactive Table, Raw (JSON), and Graph-based visualizations.



Enrichment Pipeline Generation

Export your enrichment process into reproducible pipelines to execute the same workflow on full-scale tables.



Tabular Data Annotation/Manipulation: Recap

- Semantic Table Interpretation
 - Annotation with KGs: schema vs instances
 - Entity reconciliation / CEA: a key task, e.g., W3C Reconciliation Service APIs
- LMs for Tabular Data
 - Towards generalization of tabular data understanding
 - STI + other tasks
- LLMs for Tabular Data
 - Generative approaches show strong generalization capabilities across tasks
 - Smaller models can achieve good performance if fine-tuned, top-tier models may be too slow/costly for some tasks
 - Uncertainty estimation is an interesting challenge to apply LLMs to semantic matching tasks